# A Vision Based Framework for Stress Detection in Programming Using Eyebrow Features.

Ms. Archana C
Dept. of Computer Science
School of Computing
Amrita Vishwa Vidyapeetham
Mysuru, India
archanac@my.amrita.edu

Shreyas M
Dept. of Computer Science
School of Computing
Amrita Vishwa Vidyapeetham
Mysuru, India
shreyasshre2005@gmail.com

Sohan Ponnappa P M
Dept. of Computer Science
School of Computing
Amrita Vishwa Vidyapeetham
Mysuru, India
sohanponnappa@gmail.com

Gowtham B J
lockA Dept. of Computer Science
School of Computing
Amrita Vishwa Vidyapeetham
Mysuru, India
gogowtham728@gmail.com

Harish C R
Dept. of Computer Science
School of Computing
Amrita Vishwa Vidyapeetham
Mysuru, India
harishcr389@gmail.com

Mr. Akshay S
Dept. of Computer Science
School of Computing
Amrita Vishwa Vidyapeetham
Mysuru, India
$s_akshay@my.amrita.edu$

***Abstract*—Programming tasks often impose significant cognitive load on students, leading to increased stress that can affect learning outcomes and performance. This paper presents a privacy-preserving stress detection framework that focuses on eyebrow region analysis extracted from classroom video data, thereby avoiding the use of full facial information. The proposed system follows a two-stage classification approach, where the first stage identifies stress presence and the second stage categorizes stress intensity into low, moderate, and high levels. Both traditional machine learning models and deep learning architectures are evaluated, with deep learning models demonstrating superior performance. The results indicate that the proposed approach enables efficient and real-time stress monitoring in educational environments while maintaining privacy and computational efficiency.**

***Index Terms*—Stress recognition, Facial feature extraction, Eyebrow-based analysis, Deep learning, Machine learning, Human-computer interaction**

## I. Introduction

Programming is widely recognized as a cognitively demanding activity that can lead to significant stress among students, affecting their concentration, problem-solving ability, and overall academic performance [1]. Tasks such as debugging, understanding complex logic, and working under time constraints increase mental workload and cognitive strain, which can negatively impact learning outcomes [7]. Traditional stress detection methods, including self-reported assessments and physiological measurements, have several limitations. Self-reports are subjective and may not accurately reflect real-time stress levels, while physiological approaches often require intrusive sensors and controlled environments, making them impractical for classroom use [12]. With the rise of digital learning environments, there is a growing need for automated and non-intrusive stress monitoring systems [3]. Computer vision-based approaches provide a practical solution by enabling contactless analysis of visual cues such as facial expressions and eye movements without the need for wearable devices [6]. These methods are more suitable for real-time educational settings due to their scalability and ease of deployment.

Recent research highlights that specific facial regions, particularly the eye and eyebrow areas, are strong indicators of stress and cognitive load [9]. Subtle changes in eyebrow movement and eye tension can reflect mental effort and emotional states. Focusing on these localized regions not only reduces computational complexity but also improves privacy by avoiding full-face analysis [13].

In this work, a two-stage vision-based framework is proposed for stress detection using eyebrow features extracted from classroom recordings. The first stage identifies whether a student is stressed, while the second stage classifies stress levels as low, moderate, or high. The framework combines both machine learning and deep learning techniques to achieve accurate and efficient stress recognition, while maintaining privacy and suitability for real-time applications. In addition to academic impact, stress during programming can also affect students' emotional well-being and long-term interest in the field. Continuous exposure to high cognitive load may lead to frustration, reduced confidence, and decreased engagement in learning activities. Therefore, early detection of stress can help educators provide timely support and create more adaptive and effective learning environments.

Furthermore, recent advancements in lightweight deep learning models and edge computing have made it feasible to deploy real-time stress detection systems in practical settings [14], [16]. These systems can operate efficiently with limited computational resources while maintaining high accuracy. By

combining privacy-preserving feature extraction with efficient models, stress monitoring can be seamlessly integrated into modern educational systems without disrupting the learning process.

## II. Related Work

Prior research has extensively explored stress and cognitive load in programming environments. Eye-tracking studies have shown that programming complexity and the level of expertise significantly influence cognitive load patterns [7]. In particular, increased fixation duration and the presence of complex code structures are associated with higher mental workload and stress levels [8].

Recent advancements in computer vision and deep learning have enabled effective stress detection using facial and eye-region features [9], [10]. These approaches analyze visual cues such as facial expressions, gaze patterns, and micro-expressions to estimate stress in a non-intrusive manner. However, many existing methods rely on full-face analysis or physiological signals, which may raise privacy concerns and increase computational complexity.

Despite the progress in this field, limited research has focused specifically on the eyebrow region, even though it plays a significant role in expressing emotional and cognitive states. Subtle variations in eyebrow movement and tension can provide valuable information about stress levels. This gap highlights the need for a more focused and privacy-preserving approach.

In this work, a vision-based framework is developed that utilizes eyebrow-region features for stress detection. Both traditional machine learning models and deep learning techniques are employed to analyze these features. A two-stage classification approach is adopted, where the first stage identifies the presence of stress and the second stage classifies stress levels into low, moderate, and high categories. The performance of the proposed approach is evaluated using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrices, which are widely used in stress detection research.

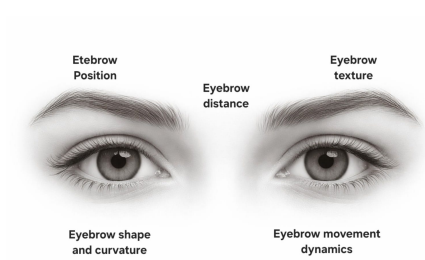


Fig. 1. **Grayscale eye image showing visual features used for stress detection.**

This section presents the machine learning and deep learning algorithms employed in the proposed stress detection and stress level analysis framework, along with their mathematical foundations. Machine learning and deep learning techniques have been widely used for stress detection and facial feature analysis due to their effectiveness in classification and pattern recognition tasks.

### A. Support Vector Machine (SVM)

Support Vector Machine aims to find an optimal hyperplane that maximizes the margin between different classes. Given a training dataset $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$, the optimization problem is defined as:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i \tag{1}$$

subject to:

$$y_i(w \cdot x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \tag{2}$$

In this work, the Radial Basis Function (RBF) kernel is used:

$$K(x_i, x_j) = \exp(-\gamma\|x_i - x_j\|^2) \tag{3}$$

### B. Random Forest (RF)

Random Forest is an ensemble learning method that combines multiple decision trees for classification. Each tree produces a class prediction, and the final output is obtained through majority voting, which improves classification accuracy and reduces overfitting.

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \ldots, T_N(x)\} \tag{4}$$

where $T_k(x)$ represents the prediction of the $k^{th}$ decision tree. Random feature selection and bootstrapped sampling improve generalization and reduce overfitting.

### C. Gradient Boosting Classifier (GBC)

Gradient Boosting builds an additive model by sequentially minimizing a loss function. The model prediction at iteration $m$ is given by:

$$F_m(x) = F_{m-1}(x) + \alpha h_m(x) \tag{5}$$

where $h_m(x)$ is the weak learner trained on the negative gradient of the loss function, and $\alpha$ is the learning rate.

### D. Extreme Gradient Boosting (XGBoost)

XGBoost extends gradient boosting by introducing regularization. The objective function is defined as:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{6}$$

where the regularization term is:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda\|w\|^2 \tag{7}$$

Here, $T$ represents the number of leaves, and $w$ denotes leaf weights.

### E. Convolutional Neural Network (CNN)

A CNN consists of convolutional, pooling, and fully connected layers. The convolution operation is defined as:

$$h_{i,j}^{(k)} = \sum_{m,n} x_{i+m,j+n} \cdot w_{m,n}^{(k)} + b^{(k)} \tag{8}$$

where $w^{(k)}$ is the convolution kernel and $b^{(k)}$ is the bias. The Rectified Linear Unit (ReLU) activation function is applied as:

$$f(x) = \max(0, x) \tag{9}$$

The final classification is obtained using the Softmax function:

$$P(y = j|x) = \frac{e^{z_j}}{\sum_k e^{z_k}} \tag{10}$$

### F. MobileNetV2

MobileNetV2 employs depthwise separable convolutions to reduce computational cost. The depthwise convolution is defined as:

$$Y_{k,l,m} = \sum_{i,j} X_{k+i,l+j,m} \cdot W_{i,j,m} \tag{11}$$

followed by pointwise convolution:

$$Z_{k,l,n} = \sum_{m} Y_{k,l,m} \cdot W_{m,n} \tag{12}$$

MobileNetV2 also uses inverted residual blocks and linear bottlenecks to improve computational efficiency while preserving representational power, making it suitable for real-time stress detection applications.

## III. Proposed Methodology

A dedicated dataset was created for this study to analyze stress levels in students during programming activities. The data was collected by recording 38 undergraduate students while they performed programming tasks in a controlled classroom environment. Each participant contributed a video of approximately 2–3 minutes, capturing natural facial expressions during problem-solving.

The recorded videos were segmented into individual frames to generate a large collection of facial images. From these frames, facial landmark detection techniques were applied to identify and extract the eyebrow region, which is the primary area of interest in this study. This approach ensures that only relevant features related to stress and cognitive load are captured while preserving user privacy.

To improve data quality, low-resolution and unclear images were removed during preprocessing. The remaining images were resized and normalized to ensure consistency across the dataset. This preprocessing step helps improve the performance and stability of both machine learning and deep learning models.

The final dataset consists of:

- **38 video recordings** collected from participants
- **4,278 facial image frames** extracted from the videos
- **4,278 corresponding eyebrow-region images**

The dataset was divided into training and testing sets to evaluate model performance. This dataset forms the foundation for all experiments conducted in this study and is specifically designed for stress detection and stress level classification in programming environments.

The proposed method uses a two-stage approach to detect stress and determine stress levels by analyzing eyebrow features of students during programming tasks. Video recordings of students performing programming activities in a classroom environment are captured and divided into individual frames. Eyebrow regions are then extracted from each frame using facial landmark detection techniques, and these eyebrow images are used for training and testing the stress detection system.

In Stage 1, eyebrow images are classified as stressed or non-stressed using traditional machine learning models such as Support Vector Machine and Random Forest as well as deep learning models including Convolutional Neural Networks and MobileNetV2.

In Stage 2, only the stressed samples are further classified into low, medium, and high stress levels. Model performance is evaluated using standard metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis, which are commonly used in stress detection studies.

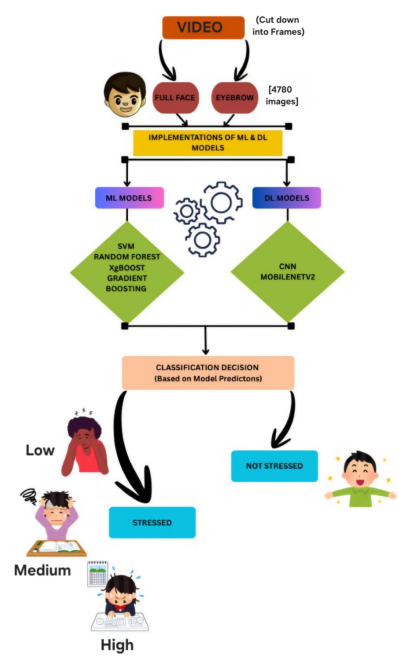


Fig. 2. **Architecture of the Proposed Stress Detection and Stress level Analysis Framework**

The proposed stress detection and stress level analysis system operates in a sequential manner. The process begins with recording videos of students while they perform programming

tasks in a classroom environment. The recorded videos are divided into individual frames, and each frame is processed to detect the face region. Facial landmark detection techniques are then applied to identify the exact location of the eyebrow region in each image [5]. This allows the system to extract stress-related information specifically from the eyebrow area. The extracted eyebrow images are preprocessed through resizing and normalization to make them suitable for model training and testing [6]. The system follows a two-stage classification framework. In the first stage, eyebrow images are analyzed to determine whether the student is stressed or non-stressed using both machine learning and deep learning approaches [11]. In the second stage, the images identified as stressed are further classified into three categories: low stress, medium stress, and high stress. The final output of the system provides information about both the presence of stress and its severity, enabling detailed assessment of student stress during programming activities. Computer-vision-based stress detection systems provide a practical and non-intrusive solution for monitoring stress in learning environments [12].

- Collected real-time video recordings of students during programming sessions and extract eyebrow region images from video frames for stress analysis.
- Developed a non-intrusive stress detection framework based on eyebrow-focused visual features.
- Designed a two-stage classification approach that first identifies the presence of stress and then categorizes stress levels as low, medium, or high.
- Implemented and evaluated traditional machine learning models, including Support Vector Machine (SVM), Random Forest, Gradient Boosting, and XGBoost, for stress detection and classification.
- Analyzed and compared the performance of deep learning models, such as Convolutional Neural Networks (CNN) and MobileNetV2, for eyebrow-based stress recognition.
- Evaluated model performance using standard metrics, including accuracy, precision, recall, F1-score, and confusion matrices.

## IV. Results and Analysis

The performance of the proposed stress detection framework is evaluated using two experimental settings: (i) eyebrow-based stress recognition and (ii) full-face stress recognition. Vision-based stress detection methods using facial features have been widely studied and provide reliable performance in stress classification tasks [5], [13]. Both approaches follow a two-stage classification strategy, where Stage 1 identifies stress presence and Stage 2 determines stress intensity levels as low, medium, or high using machine learning and deep learning techniques [11].

The performance of deep learning models was evaluated for both binary and multi-class stress classification tasks. MobileNetV2 consistently outperformed the baseline CNN in both stages, achieving higher accuracy and more balanced precision–recall scores. Deep learning models have been shown

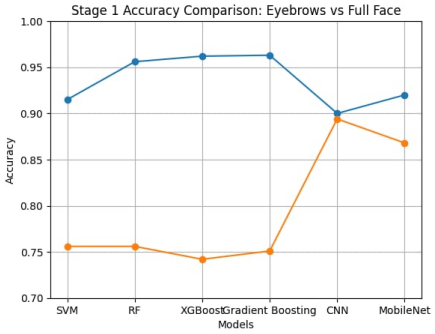


Fig. 3. **Stage 1 Accuracy Comparison:Eyebrows vs Full Face**

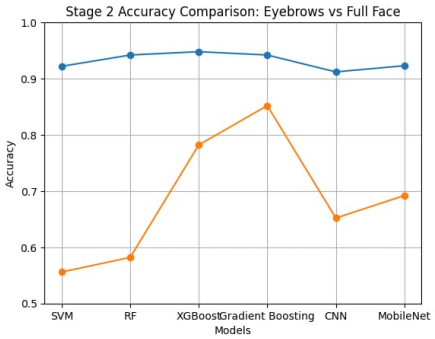


Fig. 4. **Stage 2 Accuracy Comparison:Eyebrows vs Full Face**

to provide improved performance in image-based stress classification tasks compared to traditional approaches [6]. This improvement can be attributed to transfer learning, which enables MobileNetV2 to leverage pre-trained feature representations and better capture subtle visual variations. Lightweight deep learning architectures have been demonstrated to be effective for real-time stress detection while maintaining computational efficiency [14]. The results demonstrate that lightweight transfer learning models are suitable for privacy-preserving, real-time stress detection applications [9].

### A. *Eyebrow-Based Stress Recognition Results*

TABLE I
Stage 1 Model Performance Comparison (Stressed vs Not Stressed)

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| SVM | 0.915 | 0.93 | 0.91 | 0.92 |
| Random Forest | 0.956 | 0.96 | 0.96 | 0.96 |
| XGBoost | 0.961 | 0.96 | 0.96 | 0.96 |
| Gradient Boosting | 0.961 | 0.96 | 0.96 | 0.96 |

TABLE II
STAGE 2 MODEL PERFORMANCE COMPARISON (LOW, MEDIUM, HIGH STRESS)

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| SVM | 0.920 | 0.92 | 0.92 | 0.92 |
| Random Forest | 0.941 | 0.94 | 0.94 | 0.94 |
| XGBoost | 0.947 | 0.95 | 0.95 | 0.95 |
| Gradient Boosting | 0.940 | 0.94 | 0.94 | 0.94 |

TABLE III
STAGE 1 DEEP LEARNING MODEL PERFORMANCE COMPARISON

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| CNN | 0.90 | 0.91 | 0.90 | 0.90 |
| MobileNetV2 | 0.92 | 0.92 | 0.92 | 0.91 |

TABLE IV
STAGE 2 DEEP LEARNING MODEL PERFORMANCE COMPARISON

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| CNN | 0.91 | 0.91 | 0.90 | 0.90 |
| MobileNetV2 | 0.92 | 0.92 | 0.92 | 0.92 |

The eyebrow-based approach analyzes eyebrow regions extracted from videos of students during programming tasks. Machine learning models such as SVM, Random Forest, Gradient Boosting, and XGBoost were trained using hand-crafted features, while deep learning models such as CNN and MobileNetV2 automatically learned features from eyebrow images [6]. Eyebrow and eye-region features provide useful information for stress recognition tasks [9].

In Stage 1, the models performed well in distinguishing stressed and non-stressed samples. Ensemble models such as Random Forest achieved strong classification performance, while deep learning models effectively learned stress-related eyebrow features. MobileNetV2 provided good accuracy with lower computational cost [14].

In Stage 2, stressed samples were further classified into low, medium, and high stress levels. The results indicate that eyebrow features contain sufficient information for stress level estimation, with deep learning models achieving better performance than traditional methods [10].

### B. Full-Face Stress Recognition Results

TABLE V
STAGE 1 ULTRA BASELINE MODEL PERFORMANCE COMPARISON

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| SVM | 0.756 | 0.57 | 0.76 | 0.65 |
| Random Forest | 0.756 | 0.57 | 0.76 | 0.65 |
| XGBoost | 0.742 | 0.70 | 0.74 | 0.71 |
| Gradient Boosting | 0.751 | 0.73 | 0.75 | 0.74 |

TABLE VI
STAGE 2 ULTRA BASELINE MODEL PERFORMANCE COMPARISON

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| SVM | 0.555 | 0.34 | 0.55 | 0.42 |
| Random Forest | 0.581 | 0.36 | 0.58 | 0.45 |
| XGBoost | 0.781 | 0.79 | 0.78 | 0.77 |
| Gradient Boosting | 0.852 | 0.85 | 0.85 | 0.85 |

TABLE VII
STAGE 1 PERFORMANCE COMPARISON ON FULL-FACE IMAGES

| Model | Accuracy (%) | Precision | Recall | F1-score |
|---|---|---|---|---|
| CNN | 89.27 | 0.85 | 0.86 | 0.86 |
| MobileNetV2 | 86.83 | 0.93 | 0.73 | 0.78 |

TABLE VIII
STAGE 2 PERFORMANCE COMPARISON ON FULL-FACE IMAGES

| Model | Accuracy (%) | Precision | Recall | F1-score |
|---|---|---|---|---|
| CNN | 65.16 | 0.49 | 0.64 | 0.54 |
| MobileNetV2 | 69.03 | 0.73 | 0.72 | 0.70 |

The full-face stress recognition approach analyzes multiple facial regions including the eyes, eyebrows, mouth, jaw, and forehead to detect stress. Using full-face features provides more comprehensive stress-related information and improves recognition accuracy compared to using only eyebrow features [5].

In Stage 1, full-face analysis enabled more reliable classification of stressed and non-stressed samples because additional facial features provided richer information for machine learning and deep learning models [6]. Models trained on full-face images were able to better capture stress-related expressions.

In Stage 2, stress level classification showed improved performance due to the availability of additional facial tension indicators such as jaw stiffness, lip compression, and forehead texture variations. These results confirm that full-face features enhance stress level estimation in vision-based stress detection systems [14].

### C. Comparative Analysis

TABLE IX
COMPARATIVE ANALYSIS OF EYEBROW-BASED AND FULL-FACE STRESS RECOGNITION (STAGE 1)

| Approach | Model | Accuracy | Precision | F1-Score |
|---|---|---|---|---|
| Eyebrow-Based | MobileNetV2 | 0.92 | 0.92 | 0.91 |
| Full-Face | MobileNetV2 | 0.86 | 0.93 | 0.78 |

TABLE X
COMPARATIVE ANALYSIS OF EYEBROW-BASED AND FULL-FACE STRESS RECOGNITION (STAGE 2)

| Approach | Model | Accuracy | Precision | F1-Score |
|---|---|---|---|---|
| Eyebrow-Based | MobileNetV2 | 0.92 | 0.92 | 0.92 |
| Full-Face | MobileNetV2 | 0.69 | 0.73 | 0.70 |

Comparative analysis shows that eyebrow-based stress recognition provides an effective and privacy-preserving approach while requiring lower computational cost, making it

suitable for real-time educational environments [9]. Although full-face analysis provides more detailed facial information for stress detection, eyebrow-based methods still achieve reliable performance with improved efficiency [13]. Lightweight deep learning models such as MobileNetV2 provide a good balance between accuracy and computational efficiency for stress recognition tasks [14]. The proposed two-stage classification framework further improves stress level discrimination by first identifying stressed samples and then estimating stress intensity [11]. Overall, the results indicate that eyebrow features alone provide reliable stress detection, while full-face analysis serves as a performance upper bound [13].

Although the proposed framework performs effectively, it has several limitations. The dataset was collected from a single institution, which may limit the generalizability of the system to other environments. The performance of vision-based stress detection methods can also be affected by variations in lighting conditions, camera angles, head movements, and facial occlusions such as glasses [5]. Stress labeling is subjective and may introduce classification errors [2]. Furthermore, deep learning models require considerable computational resources and do not capture temporal variations across video sequences [14].

## V. Conclusion

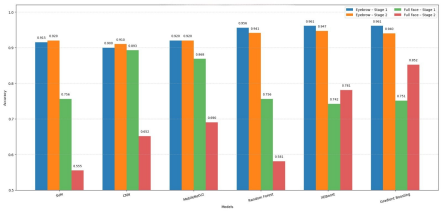


Fig. 5. **Final Conclusion Graph, Two-Stage Stress Detection:Eyebrow-Based vs Full-Face Models**

This study proposes a stress detection system for students during programming tasks using a two-stage classification approach. Students are first classified as stressed or non-stressed, and stress intensity is then categorized as low, moderate, or high using eyebrow features extracted from classroom videos [9]. The eyebrow-based approach preserves privacy and reduces computational cost while maintaining reliable performance. Compared to full-face methods, eyebrow-based models achieved consistent performance in both binary and multi-class classification tasks [13]. Deep learning models, particularly MobileNetV2, outperformed traditional classifiers, and the two-stage framework improved classification accuracy [14]. The results demonstrate that eyebrow features alone are sufficient for accurate, efficient, and privacy-preserving real-time stress monitoring in educational environments [12].

## VI. Future Work

Future work will focus on expanding the dataset to improve model generalization and exploring temporal models such as LSTM networks to capture stress patterns over time. Multimodal features including gaze behavior and additional visual cues will be integrated to improve stress detection accuracy. Real-time deployment in learning environments will also be investigated for adaptive feedback systems. Further optimization of lightweight deep learning models such as MobileNetV2 will be performed to ensure efficient and privacy-preserving stress monitoring in practical applications.


## Acknowledgment

The authors extend their heartfelt gratitude to the Department of Computer Applications (Data Science), Amrita Vishwa Vidyapeetham, Mysuru Campus, for providing the guidance, technical resources, and encouragement necessary to complete this work. Special thanks are due to Ms. Archana C for her valuable mentorship, insightful suggestions, and continuous support throughout the project's development. The authors also appreciate the participation and cooperation of peers and volunteers who contributed to dataset collection during programming sessions. Their assistance played a crucial role in shaping the success of this research.


## References


[1] M. Daudelin, et al., "Stress detection in programmers using physiological signals," 2017.
[2] N. Ruhil, et al., "Stress Detection in IT Professionals via Machine Learning," 2024.
[3] S. Sarkar, et al., "Machine learning–based real-time stress prediction using facial cues," 2020.
[4] A. B. Yadav, "Real-time Facial Emotion-based Stress Detection," 2024.
[5] M. Hosseini, et al., "Multimodal Stress Detection Using Facial Landmarks," 2023.
[6] S. Sharma and R. Dhall, "Facial Micro-Expression Based Stress Detection Using Deep Convolutional Networks," 2020.
[7] F. Viegas, et al., "Eye movement patterns during code reading and debugging," 2018.
[8] J. Kather, et al., "Plan schemata and cognitive load in software developers," 2021.
[9] M. H. Alshamrani, "Deep Learning Framework for Real-Time Stress Recognition from Eye Region," 2022.
[10] L. Wang and H. Li, "Emotion and Stress Classification using Vision Transformers," 2024.
[11] P. K. Sahu and V. Jain, "Hybrid SVM-CNN Model for Mental Stress Classification using Facial Action Units," 2021.
[12] R. T. Gomez, "Non-Intrusive Developer Stress Monitoring in IDE Environments," 2024.
[13] A. Gupta and S. Poria, "Multimodal Stress Detection Using Facial Landmarks and Heart-Rate Signals," 2023.
[14] B. R. Somani and S. Chandran, "Lightweight CNN Models for On-Device Stress Detection," 2024.
[15] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2013.
[16] A. P. K. N. Apinaya Prethi, et al., "Edge Computing Enabled Stress Detection through CNN," 2024.
[17] P. S. Gholap, et al., "IoT Enabled Stress Detection via Image Processing," 2024.
[18] Y. Zhang, et al., "Real-Time Cognitive Load Estimation in Programmers Using Eye-Tracking and Neural Networks," 2022.
[19] D. Patel, "Cognitive Stress Assessment in Software Developers using Eye Blink Rate and Facial Tension," 2023.
[20] I. McChesney and R. Bond, "Eye tracking analysis of programmers with dyslexia," 2018.